6.3 Summary

- Information theory studies the representation, quantification, and transfer of information. One of the core concepts in this field is entropy, which quantifies our uncertainty about a random variable from the variable's probability distribution.
- An entropy search procedure considers the expected reduction in entropy (and therefore in uncertainty) of a quantity of interest by taking an action, and chooses the action that maximizes this reduction. We can apply this generic procedure to many decision-making under uncertainty problems.
- Binary search may be obtained as the result of entropy search applied to the problem of finding the location of a specific number in a sorted array.
- The curse of dimensionality refers to the exponential cost of many procedures in machine learning with respect to the dimensionality of an object of interest. As the number of dimensions increases, we need exponentially more time to finish the procedure.
- In Bayesian optimization, while entropy search may be applied to the problem of finding the location of the function optimizer, it is computationally expensive due to the curse of dimensionality.
- To overcome the curse of dimensionality, we modify our goal to find the function optimized value, making it a one-dimensional search problem. The resulting Bayesian optimization policy is called Max-value Entropy Search, or MES.
- Due to the complex behavior of the global optimum of a function modeled by a Gaussian process, it's not feasible to compute the acquisition scores of MES in closed form. However, we can draw samples from probability distributions to approximate the acquisition scores.
- Implementing MES in BoTorch follows the same procedure as implementing other Bayesian optimization policies. To facilitate the sampling procedure in the acquisition score approximation, we use a Sobol sequence when initializing the policy object.

6.4 Exercise

There are two exercises in this chapter:

1. The first covers a variant of binary search in which prior information could be taken into account when making decisions.
2. The second walks us through the process of implementing Max-value Entropy Search in the hyperparameter tuning problem seen in previous chapters.

6.4.1 Incorporating prior knowledge into entropy search

We saw in Section 8.1.3 that by placing a uniform prior distribution on the location of the target within the array, the information-theoretically optimal search decision is to cut the array in half. What happens if a uniform distribution doesn't represent your prior belief faithfully and you'd like to use a different distribution? This exercise, implemented in `CH06/02 - Exercise 1.ipynb`, shows us an instance of this as well as how to derive the resulting optimal decision, and will help us further appreciate the elegance and flexibility of entropy search as a generic decision-making-under-uncertainty procedure.

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